

# Induced EMF of a Dropped Magnet

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## 1 Objective

We drop a magnet through a coil from ten different heights and read the peak-to-peak voltage  $V_{\text{peak}}$  off the scope each time. Then we plot  $V_{\text{peak}}$  against drop height  $h$  and against the magnet's speed at the coil  $v = \sqrt{2gh}$ .

## 2 Theory

Faraday's law gives the voltage induced across a coil of  $N$  turns by a changing flux  $\Phi_B$  through one turn,

$$\varepsilon = -N \frac{d\Phi_B}{dt}. \quad (1)$$

For a magnet falling along the coil axis, the flux changes faster when the magnet moves faster, so the induced voltage is proportional to the magnet's speed  $v$  at the coil.

In free fall from release height  $h$  above the coil,  $v = \sqrt{2gh}$ , so  $V_{\text{peak}} \propto v \propto \sqrt{h}$ :  $V_{\text{peak}}$  vs.  $v$  should be linear through the origin, and  $V_{\text{peak}}$  vs.  $h$  should follow a square-root curve. The biggest source of uncertainty is where we place the magnet at release, which we take as  $\pm 0.5$  cm.

## 3 Apparatus

- Aluminum support rod ( $\sim 1$  m) C-clamped to the bench; a fixture at the top anchors a vertical guide line through the coil bore so the magnet falls along the coil axis.
- Small induction coil of insulated magnet wire on a bracket mounted to the rod at bench-top height, axis vertical, leads to scope ch 1 via a BNC adapter.
- Small permanent drop magnet; same magnet, same pole down each drop.
- Oscilloscope (Tektronix TDS 1002B, 60 MHz).
- Wooden meterstick clamped to the rod, reading  $h$ ;  $\delta h = \pm 0.5$  cm from where we place the magnet.

## Circuit schematic

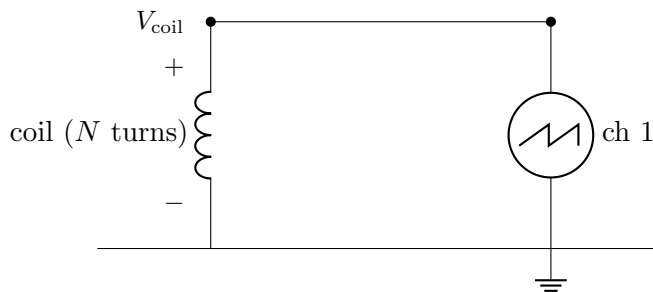


Figure 1: Coil leads connected to oscilloscope ch 1.

## 4 Procedure

1. Wire the coil to scope ch 1; set the trigger to Edge / CH1 / Rising / Normal / AC, and the time-base to 5 ms/div so the induced pulse fits on screen across the full height range (see Calculations for the  $v = \sqrt{2gh}$  and  $x/v$  estimate).
2. Clamp the meterstick vertically against the support rod with its zero at the coil top; use that reference for all ten trials.
3. Pick ten drop heights spanning roughly 10 cm to 100 cm. For each one: line up the bottom of the magnet at the target height, record  $h \pm 0.5$  cm, let it go without pushing or spinning it, and read the peak-to-peak voltage  $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$  straight off the scope display.

If a drop tumbles or the trace runs off the screen, re-take that height.

## 5 Data

We dropped the magnet from ten different heights and read  $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$  straight off the scope each time. Table 1 has one row per drop. Heights ranged from  $h \approx 0.20$  m up to about 0.96 m, capped by the guide-line clearance above the coil. We take  $\delta h = \pm 0.5$  cm from where we place the magnet, and  $\delta V_{\text{peak}} = \pm 0.04$  V from reading the scope at the V/div setting we used. Figure 2 shows the data with a  $\sqrt{h}$  curve drawn on top.

| Trial | $h$ (m)          | $V_{\text{peak}}$ (V) |
|-------|------------------|-----------------------|
| 1     | $0.20 \pm 0.005$ | $1.12 \pm 0.04$       |
| 2     | $0.30 \pm 0.005$ | $1.20 \pm 0.04$       |
| 3     | $0.35 \pm 0.005$ | $1.28 \pm 0.04$       |
| 4     | $0.40 \pm 0.005$ | $1.36 \pm 0.04$       |
| 5     | $0.47 \pm 0.005$ | $1.40 \pm 0.04$       |
| 6     | $0.55 \pm 0.005$ | $1.44 \pm 0.04$       |
| 7     | $0.65 \pm 0.005$ | $1.52 \pm 0.04$       |
| 8     | $0.80 \pm 0.005$ | $1.84 \pm 0.04$       |
| 9     | $0.87 \pm 0.005$ | $1.92 \pm 0.04$       |
| 10    | $0.96 \pm 0.005$ | $2.12 \pm 0.04$       |

Table 1: Drop height  $h$  and peak-to-peak coil voltage  $V_{\text{peak}}$  for each of the ten drops.

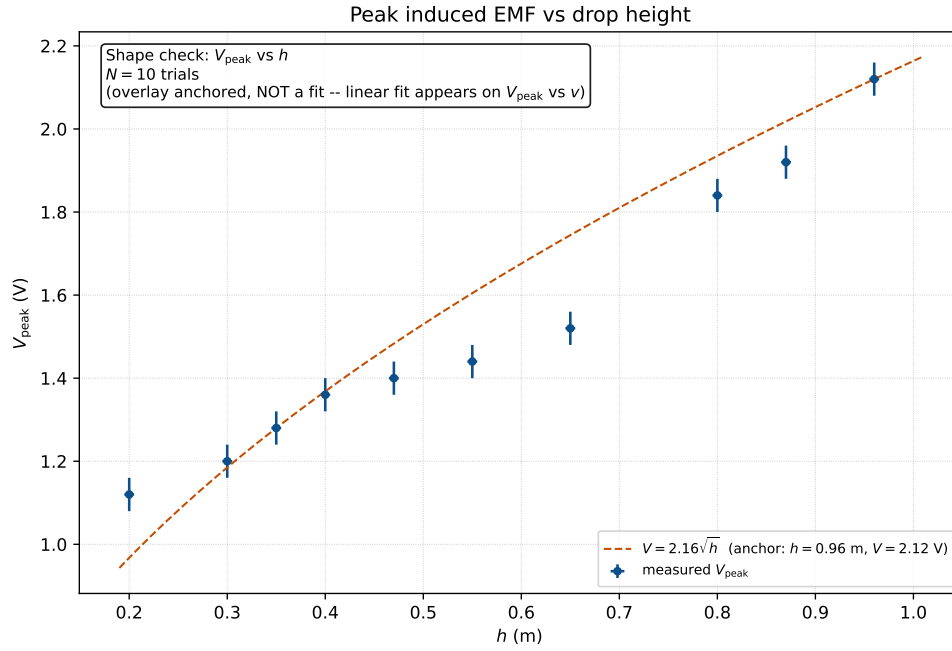


Figure 2:  $V_{\text{peak}}$  vs. drop height  $h$ , with  $\sqrt{h}$  overlay.

## 6 Calculations and Uncertainty

### Propagation rules

$$\delta(A \pm B) = \sqrt{\delta_A^2 + \delta_B^2}$$

$$\frac{\delta(AB)}{|AB|} = \frac{\delta(A/B)}{|A/B|} = \sqrt{\left(\frac{\delta_A}{A}\right)^2 + \left(\frac{\delta_B}{B}\right)^2}$$

$$\delta f = \left| \frac{df}{dx} \right| \delta x$$

### Time-base estimate (fastest drop, $h = 0.96$ m)

Before running the sweep we picked the scope's time-base from a quick energy estimate. The fastest drop is the highest one, so we use  $h = 0.96$  m to figure out how short the induced pulse can get: energy conservation gives the magnet's speed at the coil, and the coil length over that speed is roughly how long it takes the magnet to cross.

$$mgh = \frac{1}{2}mv^2 \implies v = \sqrt{2gh}$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.96} = 4.34 \text{ m s}^{-1}$$

$$t = \frac{x}{v} = \frac{0.04}{4.34} \approx \boxed{9.2 \text{ ms}}$$

We set the time-base to 5 ms/div: the fastest pulse takes about two divisions, and slower drops take a few more, so the whole signal fits on the screen.

### Velocity at the coil (Trial 6)

Inputs:  $h = 0.55 \pm 0.005$  m,  $g = 9.81 \text{ m s}^{-2}$ .

$$v = \sqrt{2gh}, \quad \delta v = \sqrt{\frac{g}{2h}} \delta h$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.55} = 3.285$$

$$\delta v = \sqrt{\frac{9.81}{2(0.55)}} \cdot 0.005 = 0.015$$

$$v = \boxed{3.29 \pm 0.02 \text{ m s}^{-1}}$$

We ran this same propagation for every row of Table 1 to get the  $v \pm \delta v$  values that go into the regression below.

**Linear regression:**  $V_{\text{peak}} = m v + b$

We fit a least-squares line to  $V_{\text{peak}}$  versus  $v = \sqrt{2gh}$  across all ten trials (Fig. 3):

$$m = \boxed{0.410 \pm 0.039 \text{ V/(m/s)}}$$

$$b = \boxed{0.203 \pm 0.128 \text{ V}}$$

$$R^2 = 0.933, \quad N = 10$$

The fit equation,  $R^2$ , and  $N$  are annotated on the figure itself.

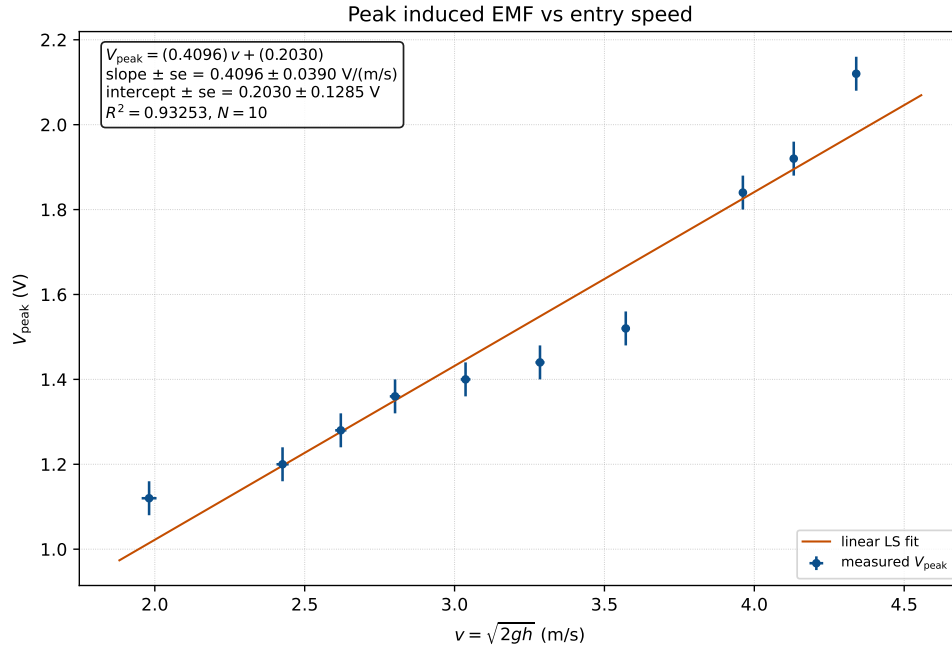


Figure 3:  $V_{\text{peak}}$  vs. coil-entry speed  $v = \sqrt{2gh}$ , with linear fit.

### **$t'$ test on the intercept**

Faraday's law says the line should go through the origin, so we expect  $b = 0$ .

$$t' = \frac{|b - 0|}{\delta b}$$

$$= \frac{0.203}{0.128}$$

$$t' = \boxed{1.58}$$

That puts  $t'$  in the borderline band ( $1 \leq t' \leq 3$ ): not quite consistent with zero, but not in outright disagreement either. So from this dataset we can't pin down whether the intercept is really zero.

## **7 Conclusion**

Over ten drops, the peak-to-peak coil voltage grew linearly with the magnet's speed at the coil  $v = \sqrt{2gh}$ , with slope  $0.41 \pm 0.04$  V/(m/s) and  $R^2 = 0.93$ . This is basically what Faraday's law predicts: a faster magnet at the coil gives a bigger induced voltage. By eye the fit line passes through or near most of the error bars; the trial at  $h \approx 0.65$  m sits visibly below the line and the highest- $h$  point sits above it, which is where most of the residual scatter comes from. The fitted intercept  $0.20 \pm 0.13$  V is small but not quite zero ( $t' = 1.58$ ); since it's only about a tenth of the  $V_{\text{peak}}$  range we swept, we read it as scope reading error rather than a real offset. The biggest source of uncertainty is the  $\pm 0.5$  cm placement of the magnet against the meterstick.